The calculations primarily describe uncollided photon attenuation under stated model assumptions. They are not a substitute for validated radiation transport simulations, professional shielding design, or regulatory radiation protection assessments.

Context

View	Complete shielding analysis
App version	1.0.0
Photon energy	1.000 MeV
Data source	srd126
Coefficient version	SRD 126; data content July 2004
Lookup method	exact tabulated
Energy bracket	1 to 1 MeV
Edge side	above
Initial primary intensity	1000 user photon units
Target transmission	0.01000 fraction

Layer 1

Material	Lead
Composition (mass fractions)	Pb:1
Composition source	https://physics.nist.gov/PhysRefData/XrayMassCoef/tab1.html
Density used	11.35 g/cm3
NIST reference density	11.35 g/cm3
Density source	https://physics.nist.gov/PhysRefData/XrayMassCoef/tab1.html
Density note	NIST reference density; some values are nominal. Not a measurement of the user sample. Change density to match sample conditions.
Physical thickness	1.000 cm
Areal density	11.35 g/cm2
Mass attenuation coefficient	0.07102 cm2/g
Linear attenuation coefficient	0.8061 1/cm
Coefficient URL	https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z82.html
Version	SRD 126; data content July 2004
Retrieval date	2026-10-14
Raw source SHA256	08e7faf3e41a0efd55cbd86a6d972c88725e04f67e7430142db37831d30272c2
Interpolation	exact tabulated
Energy bracket	1 to 1 MeV
Layer transmission	4.466e-1

Cumulative transmission	4.466e-1
Optical depth contribution	0.8061

Layer 2

Material	Water, Liquid
Composition (mass fractions)	H:0.111898; O:0.888102
Composition source	https://physics.nist.gov/PhysRefData/XrayMassCoef/tab2.html
Density used	1.000 g/cm3
NIST reference density	1.000 g/cm3
Density source	https://physics.nist.gov/PhysRefData/XrayMassCoef/tab2.html
Density note	NIST reference density; some values are nominal. Not a measurement of the user sample. Change density to match sample conditions.
Physical thickness	10.00 cm
Areal density	10.00 g/cm2
Mass attenuation coefficient	0.07072 cm2/g
Linear attenuation coefficient	0.07072 1/cm
Coefficient URL	https://physics.nist.gov/PhysRefData/XrayMassCoef/ComTab/water.html
Version	SRD 126; data content July 2004
Retrieval date	2026-10-14
Raw source SHA256	11cbc0a9d4d310a6236cf11254708a63c485e7ce606bd1e11ec175e5bbec3725
Interpolation	exact tabulated
Energy bracket	1 to 1 MeV
Layer transmission	4.930e-1
Cumulative transmission	2.202e-1
Optical depth contribution	0.7072

Primary material

Mass coefficient	0.07102 cm2/g
Mass energy-absorption coefficient (not attenuation)	0.03654 cm2/g
Linear coefficient	0.8061 1/cm
Transmission	4.466e-1
Log transmission	-0.8061
Attenuation fraction	0.5534
Attenuation factor	2.239e0

Uncollided output intensity	4.466e2 same photon units as input
Required thickness	5.713 cm
HVL	0.8599 cm
TVL	2.857 cm
Mean free path	1.241 cm
First interaction probability within thickness	0.5534

Stack

Total transmission	2.202e-1
Log transmission	-1.513
Total thickness	11.00 cm
Uncollided stack output intensity	2.202e2 same photon units as input

XCOM interactions

Independent XCOM source	https://physics.nist.gov/cgi-bin/Xcom/xcom3_1
Version	SRD 8 online response snapshot; response provides no release identifier
coherent	0.002991 cm2/g
coherent share of partial sum	0.04211 fraction
incoherent	0.04993 cm2/g
incoherent share of partial sum	0.7030 fraction
photoelectric	0.01810 cm2/g
photoelectric share of partial sum	0.2549 fraction
pairNuclear	0 cm2/g
pairNuclear share of partial sum	0 fraction
pairElectron	0 cm2/g
pairElectron share of partial sum	0 fraction
Partial coefficient sum	0.07102 cm2/g
XCOM reference total	0.07102 cm2/g
Relative residual	0.00001408
Diagnostic tolerance	0.002000
Interpretation	Fractions are normalized to the sum of partial coefficients. Residual compares that sum with the separately tabulated/interpolated XCOM total; rounding and interpolation need not commute. Tolerance is a diagnostic, not an uncertainty estimate.

Spectrum

Photon fluence ratio	1.101e-1
Energy fluence ratio	2.002e-1
Input mean photon energy	0.5500 MeV
Output mean photon energy	1.000 MeV
Duplicate entries merged	0

Spectrum line 1

Energy	0.1000 MeV
Input photon weight	1.000
Transmission	8.059e-29
Log transmission	-64.69
Output photon weight	8.059e-29

Spectrum line 2

Energy	1.000 MeV
Input photon weight	1.000
Transmission	2.202e-1
Log transmission	-1.513
Output photon weight	2.202e-1

Uncertainty

Density standard uncertainty (primary)	0.02000 g/cm3
Thickness standard uncertainty (primary)	0.005000 cm
Partial standard uncertainty of optical depth	0.004273
Partial relative standard uncertainty of primary transmission	0.004273
data	NIST tables do not supply per-value standard uncertainties.
interpolation	Unquantified. No invented percentage is assigned.
model	Scattering buildup, secondary photons, geometry and dose conversion are not included.
assumption	Independent density/thickness standard uncertainties; first-order propagation. This is a partial budget, not total uncertainty.

Equations

Linear coefficient	mu = (mu/rho) * rho
Uncollided transmission	T = exp(-mu*x)
Inverse thickness	x = -ln(T)/mu

Characteristic layers	HVL = ln(2)/mu; TVL = ln(10)/mu; lambda = 1/mu
Layers	T = exp(-sum(mu_i*x_i))
Photon fluence ratio	sum(w_j*T_j)/sum(w_j)
Energy fluence ratio	sum(E_j*w_j*T_j)/sum(E_j*w_j)

Assumptions

Model	Homogeneous layers, monoenergetic exponential attenuation per spectral line, fixed source coefficients, independent-atom mixture rule. No geometry correction, buildup, scattered/secondary photons or dose estimate.

Warnings

Warning 1	The calculations primarily describe uncollided photon attenuation under stated model assumptions. They are not a substitute for validated radiation transport simulations, professional shielding design, or regulatory radiation protection assessments.
Warning 2	Interaction contributions use a separate XCOM dataset. They are checked against its own total, not against SRD 126. Small source differences are expected.
Warning 3	HVL, TVL and mean free path shown are reference monoenergetic properties at the selected reference energy, not effective polyenergetic values.